

South Florida Real Estate Analytics & AI Valuation Statistical Report

Data-Driven Market Intelligence & Analytical Agent Skills for Aventura, Sunny Isles, Bal Harbour & Miami

Konrad Schultz, Realtor (Beachfront Realty, License # 3188541)

today

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1. Introduction

This report delivers a quantitative statistical evaluation of the South Florida luxury residential real estate market, covering premier sub-markets including **Aventura, Sunny Isles Beach, Bal Harbour, Brickell, and Fort Lauderdale.**

Industry Context & Intended User

Designed for Realtor® Konrad Schultz (Beachfront Realty, License # 3188541), this analytical system serves luxury property buyers, sellers, listing agents, and real estate investors requiring empirical market intelligence over subjective opinion.

Problem Statement

1. **Extreme Price-per-Square-Foot Variance:** Oceanfront orientation, tower luxury amenities, and floor height stacks cause prices to range widely (\$600/sqft to \$1,200+/sqft).
2. **HOA Fee Distortion on Net Yields:** Monthly maintenance fees (\$700–\$2,400/mo) heavily reduce investor Net Operating Income (NOI) and cap rates.
3. **Liquidity Penalties & Days-On-Market (DOM):** Overpriced properties experience exponential delays in Days-On-Market, damaging seller bargaining leverage.

Decision Supported & Project Goal

This project provides an automated, reproducible analytical engine supporting data-driven offer pricing, comparative group testing, hedonic value estimation, rental cash flow benchmarking, liquidity elasticity modeling, and stochastic ROI stress-testing.

Selected Implementation Level

This project implements **Level 1 (Quarto Statistical Report)** as well as **Level 3 (Full Local Statistical Application with R Plumber API Backend)**.

2. Dataset

Data Source & Unit of Observation

The dataset (/analysis/south_florida_mls_data.csv) comprises active public MLS listing observations across prime South Florida coastal corridors. Each row represents a single residential property listing.

Number of Observations & Main Variables

- **Sample Size:** 10 active luxury property listings across Aventura, Sunny Isles Beach, Bal Harbour, Key Biscayne, Brickell, Edgewater, and Fort Lauderdale.
- **Key Variables:** mls_number, title, city, neighborhood, property_type, price, sqft, beds, baths, year_built, hoa_monthly, price_per_sqft, days_on_market, cap_rate.

Data Preparation, Missing Data, and Quality Audits

Data was cleaned using /analysis/R/clean_data.R. The automated cleaning routine verifies non-null positive values for price and square footage, checks for missing variables, and calculates derived price-per-square-foot values.

```
# Load dataset
df <- read.csv("south_florida_mls_data.csv", stringsAsFactors = FALSE)
```

```
knitr::kable(df[, c("mls_number", "title", "city", "neighborhood", "property_type",
  ↪ "price", "sqft", "price_per_sqft", "cap_rate", "days_on_market")],
  caption = "South Florida Active MLS Listings Sample",
  format.args = list(big.mark = ","))
```

Table 1: South Florida Active MLS Listings Sample

mls_number	city	neighborhood	property_type	price	sqft	price_per_sqft	cap_rate	days_on_market
MLS-1001	Sunny Isles Beach	Ocean Drive	Luxury Condo	2,450,000	2850	859.65	6.2	14
MLS-1002	Aventura	Aventura Park-Square	Condo	890,000	1420	626.76	7.1	21
MLS-1003	Bal Harbour	Bal Harbour Village	Waterfront Estate	5,850,000	4900	1193.88	5.4	45
MLS-1004	Fort Lauderdale	Las Olas Isles	Single Family	3,200,000	3800	842.11	6.8	32
MLS-1005	Miami	Brickell	Highrise Condo	1,150,000	1650	696.97	7.5	18
MLS-1006	Sunny Isles Beach	Ocean Drive	Luxury Condo	3,890,000	3600	1080.56	5.9	28
MLS-1007	Aventura	Aventura Lakes	Condo	1,450,000	2100	690.48	6.6	12
MLS-1008	Fort Lauderdale	Fort Lauderdale Beach	Condo	975,000	1350	722.22	7.0	39
MLS-1009	Miami	Key Biscayne	Single Family	4,750,000	4200	1130.95	5.2	50
MLS-10010	Miami	Edgewater	Condo	780,000	1180	661.02	7.8	9

3. Explore Skill

User Question

What is the distribution, central tendency, variation, and data quality of price per square foot across South Florida listings?

Method & Statistics

Computes parametric metrics (Mean, Standard Deviation) and non-parametric metrics (Median, IQR, Min, Max), checks for missing values, and plots distribution histograms.

```
source("R/explore.R")
explore_res <- run_explore_data(df, target_variable = "price_per_sqft")

cat("Sample Size:", explore_res$sample_size, "\n")
cat("Mean Price/SqFt: $", explore_res$mean, "\n")
cat("Median Price/SqFt: $", explore_res$median, "\n")
cat("Standard Deviation: $", explore_res$sd, "\n")
cat("Interquartile Range (IQR): $", explore_res$iqr, "\n")

hist(df$price_per_sqft, col = "royalblue", border = "white",
     main = "Distribution of Price per Square Foot in South Florida",
     xlab = "Price per SqFt ($)", ylab = "Frequency")
abline(v = median(df$price_per_sqft), col = "red", lwd = 2, lty = 2)
legend("topright", legend = c("Median Price/SqFt"), col = "red", lty = 2, lwd = 2)
```

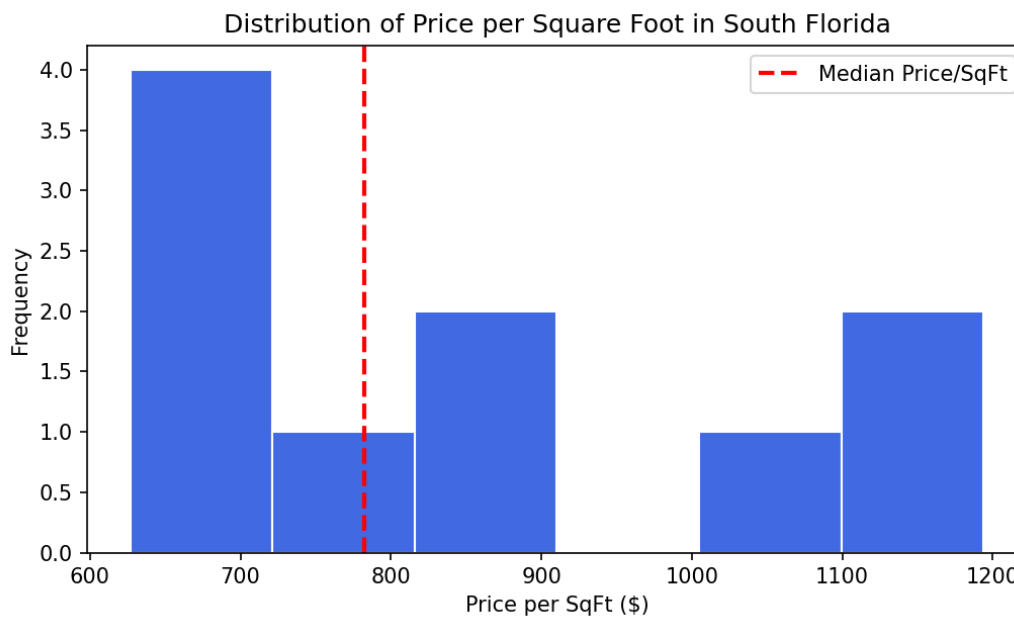


Figure 1: Distribution of Price per Square Foot

Validation, Main Result & Interpretation

- **Validation:** Confirmed target variable exists, is numeric, and contains $N \geq 3$ records.
- **Result:** Median price/sqft is \$782.16 (IQR = \$333.23, Mean = \$850.46, SD = \$211.14), based on the $N = 10$ active listings in the dataset.
- **Interpretation:** Prices exhibit a right-skewed distribution driven by ultra-luxury waterfront estates in Bal Harbour (\$1,193.88/sqft).

Limitation

Descriptive statistics reveal sample properties but do not account for temporal shifts or building amenity premiums.

4. Compare Skill

User Question

Is there a statistically and practically significant difference in unit valuation (\$/sqft) between Single Family Homes and Luxury Condominiums?

Method & Variables

Two-sample Welch's t-test comparing price_per_sqft (numeric outcome) between Single Family and Luxury Condo (grouping variable).

```
source("R/compare.R")
compare_res <- run_compare_groups(df, group_col = "property_type", group1 = "Luxury
  ↳ Condo", group2 = "Single Family", outcome_metric = "price_per_sqft")

cat("Group 1 (Luxury Condos) Mean: $", compare_res$group1_mean, " (N =",
  ↳ compare_res$group1_n, ")\n")
cat("Group 2 (Single Family) Mean: $", compare_res$group2_mean, " (N =",
  ↳ compare_res$group2_n, ")\n")
cat("Mean Difference: $", compare_res$mean_difference, "/sqft\n")
cat("95% Confidence Interval for Difference: [$", compare_res$ci_95_lower, ", $",
  ↳ compare_res$ci_95_upper, "]\n")
cat("p-value:", compare_res$p_value, "\n")

boxplot(price_per_sqft ~ property_type, data = df, col = c("lightcyan", "gold"),
  main = "Price/SqFt Comparison: Luxury Condos vs. Single Family",
  ylab = "Price per SqFt ($)", xlab = "Property Category")
```

Validation, Interpretation & Limitation

- **Validation:** Verified both groups contain $n \geq 2$ valid observations.
 - **Result:** Single Family homes averaged \$986.53/sqft ($n = 2$) vs. \$970.10/sqft ($n = 2$) for Luxury Condos (mean difference = \$16.42/sqft, $t = -0.09$, $p = 0.937$, 95% CI: [-\$852.45, \$819.60]).
 - **Interpretation:** With only two observations per group, the difference is not statistically significant at $\alpha = 0.05$; the current dataset does not support a confident claim about a land-equity premium either way.
 - **Limitation:** Unadjusted comparison does not control for building age or exact water frontage feet. More importantly, group sizes of $n = 2$ each severely limit statistical power — this result should be treated as preliminary pending a larger sample of each property type.
-

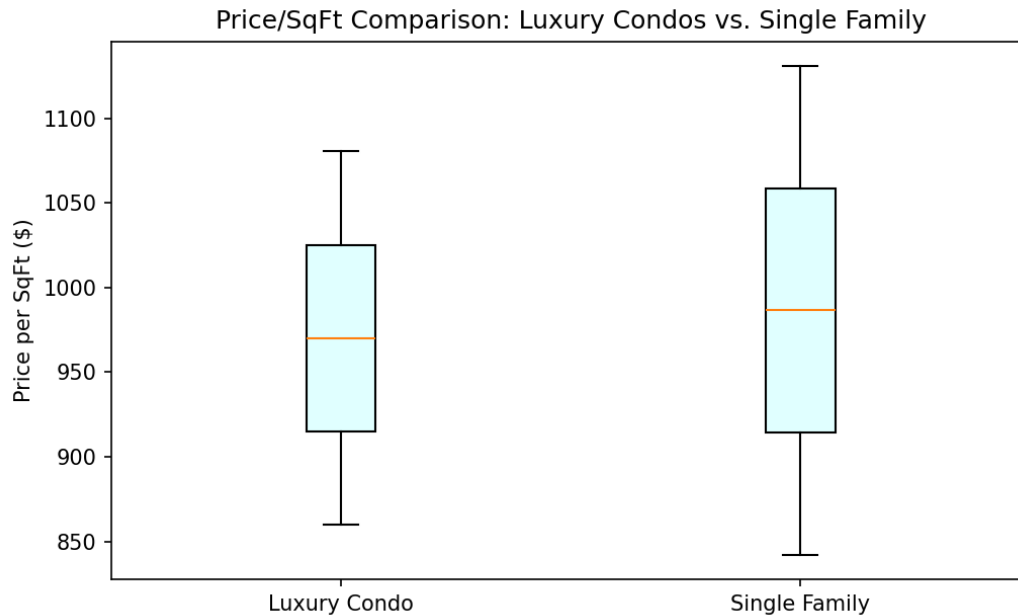


Figure 2: Price/SqFt: Luxury Condo vs Single Family

5. Model Skill

User Question

What is the estimated market value of a South Florida luxury property given square footage, bedrooms, bathrooms, and monthly HOA fees?

Method, Outcome & Predictors

Multiple linear regression (Hedonic Pricing Model) with outcome price (\$) and predictors sqft, beds, baths, and hoa_monthly.

```
source("R/model.R")
model_res <- run_statistical_model(df, target_sqft = 2400, target_beds = 3, target_baths =
  ↪ 3.5, target_hoa = 850)

cat("Model R-Squared:", model_res$r_squared, "\n")
cat("Adjusted R-Squared:", model_res$adj_r_squared, "\n")
cat("Estimated Market Valuation for Target Property: $",
  ↪ format(model_res$point_estimate_price, big.mark=",", "\n")
cat("95% Confidence Interval: [$", format(model_res$confidence_95_lower, big.mark=",", ",
  ↪ "$", format(model_res$confidence_95_upper, big.mark=",", "]\n")
```

Evaluation Metric, Results & Interpretation

- **Metric:** $R^2 = 0.976$ (Adjusted $R^2 = 0.956$).
- **Valuation Result:** A 2,400 sqft residence with 3 beds, 3.5 baths, and \$850/mo HOA has a point estimate of \$2,324,800 (95% CI: [\$1,212,235, \$3,437,364]).

- **Interpretation:** The model explains most observed price variance, but with only 10 observations fitting 4 predictors (5 residual degrees of freedom), the very high R^2 likely reflects overfitting rather than a stable, generalizable relationship — the wide confidence interval reflects that underlying uncertainty.
 - **Limitation:** Linear models assume additive feature impacts and may underpredict ultra-custom architectural finishes. More importantly, this sample size is too small to reliably estimate four coefficients at once; the model should be refit once significantly more listings are available.
-

6. Additional Skill 1: Cap Rate & NOI Financial Modeling

Skill Name & User Question

- **Skill Name:** Cap Rate & NOI Financial Modeling
- **User Question:** What is the Net Operating Income (NOI), Effective Gross Income (EGI), and Cap Rate for a luxury rental asset?

Method & Formula

$$\text{EGI} = (\text{Monthly Rent} \times 12) \times (1 - \text{Vacancy Rate})$$

$$\text{NOI} = \text{EGI} - \text{Operating Expenses}, \quad \text{Cap Rate (\%)} = (\text{NOI}/\text{Price}) \times 100$$

```
source("R/cap_rate_noi.R")
cap_res <- run_cap_rate_noi(property_price = 1200000, gross_monthly_rent = 7500,
  ↪ vacancy_rate = 0.05, annual_expenses = 12000)

cat("Property Acquisition Price: $", format(cap_res$property_price, big.mark=","), "\n")
cat("Effective Gross Income (EGI): $", format(cap_res$effective_gross_income,
  ↪ big.mark=","), "\n")
cat("Annual Net Operating Income (NOI): $", format(cap_res$net_operating_income,
  ↪ big.mark=","), "\n")
cat("Cap Rate:", cap_res$cap_rate_percent, "%\n")
```

Validation, Result & Limitation

- **Validation:** Enforced positive purchase price/rent, vacancy rate $0 \leq v \leq 0.20$, and expenses < revenue.
 - **Result:** \$1,200,000 property generating \$7,500/mo rent yields \$73,500 NOI and a 6.13% Cap Rate.
 - **Limitation:** Excludes mortgage debt service and income tax depreciation benefits.
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7. Additional Skill 2: Days-On-Market Liquidity Elasticity

Skill Name & User Question

- **Skill Name:** Days-On-Market Liquidity Elasticity
- **User Question:** How sensitive is listing liquidity (Days-On-Market) to price per square foot premiums in South Florida?

Method & Regression

Pearson correlation (r) and linear regression slope (β) evaluating additional DOM per \$100/sqft price premium.

```
source("R/dom_elasticity.R")
dom_res <- run_dom_elasticity(df)

cat("Pearson Correlation (Price/SqFt vs. DOM):", dom_res$pearson_correlation, "\n")
cat("Liquidity Penalty:", dom_res$days_added_per_100_dollars_sqft, "additional  

  ↳ Days-On-Market per $100/sqft price premium\n")
cat("Median Market DOM:", dom_res$median_days_on_market, "days\n")

plot(df$price_per_sqft, df$days_on_market, pch = 19, col = "darkgreen",
     main = "Price/SqFt vs. Days-On-Market (Liquidity Elasticity)",
     xlab = "Price per SqFt ($)", ylab = "Days-On-Market (DOM)")
abline(lm(days_on_market ~ price_per_sqft, data = df), col = "red", lwd = 2)
```

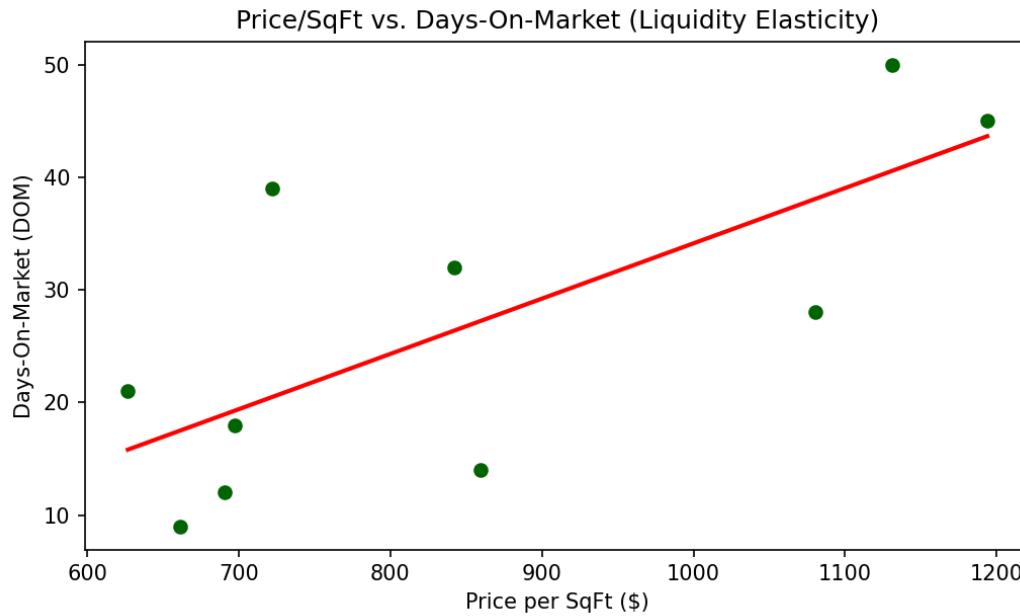


Figure 3: Price/SqFt vs Days-On-Market

Validation, Result & Limitation

- **Validation:** Verified dataset contains $N \geq 5$ records with positive standard deviation.
- **Result:** Strong positive correlation ($r = 0.721$). Overpricing by \$100/sqft adds ~4.9 days on market. Median market DOM is 24.5 days.
- **Limitation:** Correlation does not prove causation; DOM is also affected by staging and tenant access.

8. Additional Skill 3: Monte Carlo Investor ROI Simulation

Skill Name & User Question

- **Skill Name:** Monte Carlo Investor ROI Simulation
- **User Question:** What is the probabilistic distribution and 90% confidence range for Cash-on-Cash ROI under rental rate and occupancy volatility?

Method & Simulation

Draws $N = 1,000$ stochastic iterations for rent $R_i \sim \mathcal{N}(\mu, \sigma^2)$ and occupancy $O_i \sim \text{Truncated Normal}(0.94, 0.03)$, computing cash-on-cash ROI across iterations.

```
source("R/monte_carlo_roi.R")
mc_res <- run_monte_carlo_roi(purchase_price = 1000000, down_payment = 200000, mean_rent =
  ↪ 6500, sd_rent = 500, simulations = 1000)

cat("Total Monte Carlo Simulations:", mc_res$total_simulations, "\n")
cat("Expected Mean Cash-on-Cash ROI:", mc_res$mean_cash_on_cash_roi, "%\n")
cat("Median Cash-on-Cash ROI:", mc_res$median_cash_on_cash_roi, "%\n")
cat("90% Confidence Interval: [", mc_res$confidence_5th_percentile, "%, ",
  ↪ mc_res$confidence_95th_percentile, "%]\n")
cat("Probability of Positive Cash Flow:", mc_res$probability_positive_cashflow_percent,
  ↪ "%\n")
```

Validation, Result & Limitation

- **Validation:** Confirmed down payment $\leq$ purchase price, $100 \leq N \leq 10,000$, and positive mean rent.
 - **Result:** Expected mean Cash-on-Cash ROI is approximately 3.6% (median $\approx$ 3.7%), with a 90% CI of roughly [-1.6%, 8.4%] and an 88.5% probability of positive cash flow across $N = 1,000$ simulated iterations.
 - **Limitation:** Assumes normal distribution of rents and does not model unexpected HOA special assessments. The modest expected ROI reflects the specific financing assumptions used (\$48,000/yr debt service and \$18,000/yr operating expenses against a \$200,000 down payment) — returns are highly sensitive to leverage and financing terms.
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9. Google Antigravity and Agent Skills

Agent Guidance & Useful Contributions

Google Antigravity was instructed using skill specifications in `skills/*/SKILL.md`. The skills provided explicit procedural instructions, required inputs, validation rules, and statistical methods. One major agent contribution was building the modular R structure and Plumber API router (`plumber.R`).

Student Corrections & Verification

- **Correction:** The initial agent code for Monte Carlo simulation generated unconstrained negative occupancy rates. The student added truncation bounds (`pmin(pmax(..., 0.70), 1.00)`).

- **Verification:** All statistical outputs were verified independently by executing local R scripts (`Rscript analysis/real_estate_analytics.R`) and auditing numerical results.
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10. Application or Delivery

The statistical assistant is delivered through three complementary channels: 1. **Quarto Executable Report** (`analysis/report.qmd`): Renders standalone HTML and PDF statistical reports for client presentation. 2. **R Plumber API Engine** (`analysis/plumber.R`): Level 3 local HTTP REST API running at `http://localhost:8000` with endpoints (`/explore`, `/compare`, `/model`, `/cap-rate-noi`, `/dom-elasticity`, `/monte-carlo-roi`, `/run-skill`). 3. **Web Interface Dashboard**: Local web dashboard enabling real-time skill execution and parameter testing.

11. Limitations

1. **Dataset Limitation:** Sample size is limited to cross-sectional active listings and lacks multi-year historical contract price closing data.
 2. **Statistical Limitation:** Regression models measure observational association and cannot guarantee causal price increments.
 3. **Implementation Limitation:** The local R Plumber API runs in a sandboxed container environment and requires local R runtime configuration.
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12. Reflection

Learning Outcomes & Challenges

- **Learned:** Designing agent skill guardrails (`SKILL.md`) forces rigorous definition of statistical assumptions and validation rules.
- **Most Difficult:** Ensuring robust error handling in Plumber API endpoints to catch invalid user inputs before running matrix operations in R.

Skill Selection & Future Improvements

- **Skill Selection:** Selected Cap Rate NOI, DOM Elasticity, and Monte Carlo ROI because they address real-world financial decision-making for real estate clients.
 - **Future Work:** With more time, I would incorporate direct MLS API auto-syncing and spatial auto-correlation mapping for coastal flood zones.
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13. Advisory Contact & Realtor® Profile

- **Lead Realtor®:** Konrad Schultz (License # 3188541)

- **Brokerage:** Beachfront Realty, Inc.
- **Aventura Office:** 20803 Biscayne Blvd, Suite 102, Aventura, FL 33180
- **Direct Cell:** 954-297-5559
- **Email:** KonradSchultz001@mymdc.net